IHSAN ULLAH

G7/3 Islamabad • +92 340 0552325 • ihsaanullah642@gmail.com
GitHub • LinkedIn • DOB: 03-01-2002 • Nationality: Pakistani

Professional Summary

AI/ML Engineer specialized in quantitative trading systems with expertise in Vision-Language Models (VLM), Large Language Models (LLM), and algorithmic trading strategies. Proven experience building AI-powered fintech applications integrating advanced technical analysis (Elliott Wave, Fibonacci, Gann, Wyckoff, Ichimoku) with state-of-the-art deep learning models. Skilled in deploying production-grade ML infrastructure with Docker, FastAPI, and GPU-accelerated workloads. Strong foundation in reinforcement learning, NLP transformers, and quantitative financial modeling with emphasis on automated trading signal generation and market analysis.

Work Experience

AI Engineer | ByteBoom

February 2024 - December 2024

- Architected advanced Vision-Language Model (VLM) trading system integrating GPT-4 Vision and Claude Vision for multi-timeframe technical analysis, processing TradingView chart screenshots to validate trading signals through comprehensive market pattern recognition including trend detection, support/resistance levels, and candlestick formations
- Engineered quantitative trading algorithms implementing Elliott Wave theory, Fibonacci retracements, Gann analysis, Wyckoff methodology, and Ichimoku Cloud indicators for multi-dimensional market analysis, liquidity detection, and automated signal generation across cryptocurrency and forex markets
- Developed Reinforcement Learning (PPO) trading bot with Neo4j GraphDB integration, extracting complex market features from graph database to construct high-dimensional state vectors for action prediction and value estimation, optimizing trading decisions through deep RL techniques
- Implemented Large Language Model (LLM) integration for financial news sentiment analysis and market narrative extraction, combining NLP pipelines with quantitative signals to enhance trading decision accuracy
- Built VLM-powered automated trading signal confirmation system capturing multi-timeframe screenshots with technical indicators, leveraging computer vision and natural language processing for comprehensive market analysis and trade validation (Live Demo: vlm_fallback.57.128.242.183.sslip.io)

- Orchestrated production ML infrastructure with Docker containerization ensuring scalable deployment of trading models, implementing CI/CD pipelines and maintaining high-availability systems for real-time market analysis
- Led collaborative development using Git with role-based access control (RBAC), branch management strategies, and structured code review processes for maintaining robust fintech codebase

AI Engineer | University of Peshawar

November 2023 - May 2024

- Deployed production-grade NLP models using FastAPI and Docker for scalable inference serving in Linux environments, optimizing model training and inference workflows with GPU acceleration
- Fine-tuned RoBERTa transformer model achieving 95% F1-score for fake news detection, implementing advanced NLP architectures and hyperparameter optimization techniques
- Developed GRU-enhanced RoBERTa classifier for fake review detection, leveraging transfer learning techniques to improve model accuracy and generalization performance
- Optimized machine learning algorithm performance through systematic hyperparameter tuning, feature engineering, and model architecture experimentation in GPU-accelerated environments

Key Projects

VLM-Powered Quantitative Trading System

Built automated multi-timeframe chart analysis system integrating GPT-4 Vision and Claude Vision for technical pattern recognition. Implemented Elliott Wave, Fibonacci, Gann, Wyckoff, and Ichimoku indicators for comprehensive market analysis. Deployed real-time trading signal validation with computer vision and NLP. Live Demo: vlm_fallback.57.128.242.183.sslip.io

Reinforcement Learning Trading Bot with Graph Database

Implemented PPO (Proximal Policy Optimization) model with Neo4j GraphDB integration for feature extraction. Constructed complex state vectors from market graph data for action prediction and value estimation in automated trading decisions.

Quantitative Trading Signals Application

Developed comprehensive algorithmic trading system implementing Elliott Wave theory, Fibonacci retracements, Gann analysis, Wyckoff methodology, and Ichimoku Cloud for multi-dimensional market analysis, trend detection, and liquidity identification across multiple asset classes.

Fake News Detection Web Application

Built Streamlit-based ML application implementing TF-IDF, Logistic Regression, BERT, and RoBERTa with feedforward neural networks. Achieved 96% accuracy using RoBERTa + FFNN architecture. GitHub: github.com/IhsanUllah-AI/fake-news-detection

Fake Reviews Detection System

Engineered classifier combining RoBERTa transformer with GRU recurrent layers, achieving 93% accuracy for e-commerce review authenticity detection using transfer learning and ensemble methods.

Education

- BS Computer Science | University of Peshawar | May 2025
- Intermediate Pre-Engineering | Jahanzeb College Swat | July 2019

Technical Skills

Quantitative Finance & Trading: Elliott Wave Theory, Fibonacci Analysis, Gann Theory, Wyckoff Methodology, Ichimoku Cloud, algorithmic trading, technical analysis, market microstructure, quantitative strategies, signal processing

AI/ML for Finance: Vision-Language Models (GPT-4 Vision, Claude Vision), Large Language Models, sentiment analysis, market pattern recognition, predictive modeling, time-series forecasting, computer vision for trading charts

Deep Learning & NLP: Transformer models (BERT, RoBERTa, GPT), fine-tuning (LoRA, QLoRA), PyTorch, TensorFlow, LSTM, GRU, CNN, NLP pipelines, sentiment analysis, text classification

Reinforcement Learning: PPO (Proximal Policy Optimization), deep RL algorithms, policy optimization, value estimation, trading agents, multi-agent systems

ML Infrastructure & Deployment: Docker containerization, FastAPI, MLOps, GPU scheduling (CUDA), Linux systems, model serving, vLLM runtime, production deployment, CI/CD pipelines

Data Engineering: Neo4j GraphDB, feature engineering, data preprocessing, NumPy, Pandas, scikit-learn, time-series data, market data processing

Programming & Development: Python (Advanced), JavaScript, C++ (Basic), Git version control, RBAC, API development, system architecture

Visualization & Analysis: Matplotlib, Seaborn, Plotly, Streamlit, TradingView, market data visualization, technical chart analysis

Certifications

- Machine Learning Specialization
- Deep Learning Specialization

Languages

- Urdu: Fluent
- English: Professional proficiency

Volunteer Experience

Volunteer Recruitment Coordinator | University of Peshawar

January 2023 - Present

- Conducted outreach activities including job fairs, networking events, and community meetings to promote volunteer opportunities and student engagement
- Organized information sessions, open houses, and informational interviews for prospective volunteers to increase participation
- Provided guidance and support to new students including orientations on organizational policies, procedures, and program requirements